

Programme	B. Sc. Computer Science				
Course Title	Exploring Cyber security in social media				
Type of Course	Minor				
Semester	II				
Academic Level	100-199				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	3	-	2	75
Pre-requisites	1. Fundamentals of Computer science				
Course Summary	Students can investigate the complex interplay between social media and cyber security with this minor programme. The course will explore the different risks, vulnerabilities, and dangers related to social media platforms, providing participants with valuable knowledge on how to safeguard both individuals and organisations. Students will get a thorough grasp of cyber security principles as they relate to social media through a combination of academic study, hands-on activities, and case analysis.				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Understand the idea of cyber security as well as the problems and difficulties that surround it.	U	F	Instructor-created exams / Quiz
CO2	Understand the cyber crimes, their nature, legal remedies and as to how report the crimes through available platforms and procedures	U	C	Practical Assignment / Observation of Practical Skills
CO3	Understand the privacy and security issues associated with using online social media. They should also be aware of the best practices for using social media platforms, the legal ramifications, and how to report incorrect content.	U	F	Seminar Presentation / Group Tutorial Work/ Viva Voce

CO4	Understand ethical standards related to usage of social media and apply those ethical standards in their day today life usage.	U	C	Instructor-created exams / Home Assignments
CO5	Comprehend the fundamentals of computer and mobile security and will be able to safeguard their gadgets with simple tools and technology.	Ap	P	Writing assignments/ Instructor-created exams/ practicals
CO6	Develop a cybersecurity plan for a hypothetical social media.	Ap	P	Case Study/ mini Project/ practicals
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Module	Unit	Content	Hrs (45+30)	Marks (70)
I	Introduction to Cyber security & History of Internet and social media		9	12
	1	Historical overview of social media development	2	
	2	Impact of social media	1	
	3	Internet, World wide web, Introduction of the internet	2	
	4	Internet infrastructure for data transfer and governance	2	
	5	Terminologies like anti-virus, firewall, Wi-Fi network	2	
II	Introduction Cyber Security & reporting of cyber crimes		12	18
	6	Concept of cyber security, Issues and challenges of cyber security Terminologies: Cyber Security, Cyber Crime, Cyber Attack, Cyber Espionage, Cyber Warfare	2	
	7	Classification of cyber crimes : Financial crimes: Online fraud, phishing, identity theft (basic concepts only)	2	
	8	Cyber crime targeting computers and mobiles	2	
	9	Cyber crime against women and children, social engineering attacks, malware and ransomware attacks	2	
	10	Reporting of cyber crimes,	2	

	11	Legal perspective of cyber crime, IT Act 2000 and its amendments, Cyber crime and offences	2	
III	Introduction to Social Media		12	20
	12	Introduction to Social networks. Types of Social media, Social media platforms	2	
	13	Social media monitoring, Hashtag, Viral content	3	
	14	Social media marketing	2	
	15	Social media privacy, Challenges, opportunities and pitfalls in online social network	2	
	16	Security issues related to social media: Phishing Attacks, Account take over, Data breeches, Fake Accounts and Impersonation, Credential Stuffing, Doxing (concepts only)	3	
IV	Cyber Security in social media		12	20
	17	End Point device and Mobile phone security, Password policy	1	
	18	Data backup, Downloading and management of third party software	2	
	19	Cyber Security best practices, Significance of host firewall and Ant-virus, Management of host firewall and Anti-virus,	2	
	20	Wi-Fi security, Configuration of basic security policy and permissions.	1	
	21	Terminologies like- strong password, Two-Factor Authentication, Login Activity Monitoring, Authorized Devices	3	
	22	Ethical dilemmas in social media usage: Privacy vs. Transparency, Authenticity vs. Self-Presentation, Misinformation vs. Truthfulness, Cyberbullying and Online Harassment, Data Privacy and User Consent, Influence and Manipulation, Addiction & Mental Health	3	
V	Practical Implementations of Cyber security in social media		30	
	1	● Setting, configuring and managing three password policy in the computer (BIOS, Administrator and Standard User). ● Setting and configuring two factor authentication in the Mobile phone. ● Security patch management and updates in Computer and Mobiles. ● Managing Application permissions in Mobile phone. ● Installation and configuration of computer Anti-virus. ● Installation and configuration of Computer Host Firewall.	20	

		● Wi-Fi security management in computer and mobile. ● Hands-on exercises with social media monitoring tools		
	2	Develop a cybersecurity plan for a hypothetical social media scenario (Capstone) Organisations dealing with Cyber crime and Cyber security in India, Case studies.	10	

References

- "Social Media Security: Leveraging Social Networking While Mitigating Risk" by Michael Cross
- "The Social Media Security Playbook: Your Guide to Stopping Threats, Plugging Gaps, and Responding to Emergencies" by Christopher Hadnagy and Michele Fincher
- Cyber Security Understanding Cyber Crimes, Computer Forensics and Legal Perspectives by Sumit Belapure and Nina Godbole, Wiley India Pvt. Ltd. (First Edition, 2011)
- Security in the Digital Age: Social Media Security Threats and Vulnerabilities by Henry A. Oliver, Create Space Independent Publishing Platform. (Pearson , 13th November, 2001)
- Fundamentals of Network Security by E. Maiwald, McGraw Hill.

Mapping of COs with PSOs and POs :

	PSO1	PSO2	PSO3	PSO4	PSO5	PSO6	PO1	PO2	PO3	PO4	PO5	PO6
CO 1	-	2	-	-	-	-						
CO 2	-	1	-	-	-	-						
CO 3	-	2	-	-	-	-						
CO 4	-	2	-	-	-	-						
CO 5	-	1	-	-	-	-						
CO 6	-	1	1	-	-	-						

Correlation Levels:

Level	Correlation
-	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Quiz / Assignment/ Quiz/ Discussion / Seminar
- Midterm Exam
- Programming Assignments (20%)
- Final Exam (70%)

Mapping of COs to Assessment Rubrics :

	Internal Exam	Assignment	Practical Evaluation	End Semester Examinations
CO 1	✓	✓	✓	✓
CO 2	✓	✓	✓	✓
CO 3	✓	✓	✓	✓
CO 4	✓	✓	✓	✓
CO 5	✓	✓	✓	✓
CO 6	✓	✓	✓	